respectively as

a) $\left(3.25 \times 10^{-2} \text{ mol } L^{-1}s^{-1}\right)$ and $\left(1.3 \times 10^{-2} \text{ mol } L^{-1}s^{-1}\right)$

b) $\left(1.3 \times 10^{-2} \text{ mol } L^{-1}s^{-1}\right)$ and $\left(3.25 \times 10^{-2} \text{ mol } L^{-1}s^{-1}\right)$

c) $\left(1.3 \times 10^{-1} \text{ mol } L^{-1}s^{-1}\right)$ and $\left(3.25 \times 10^{-2} \text{ mol } L^{-1}s^{-1}\right)$

d) None of these

19. During the decomposition of H_2O_2 to give dioxygen, 48 g O_2 is formed per minute at certain point of time. The rate of formation of water at this point is

a) 0.75 mol min^{-1}

b) 1.5 mol min^{-1}

c) 2.25 mol min^{-1}

d) 3.0 mol min^{-1}

20. If the initial concentration of the reactant is doubled, the time for half reaction is also doubled. Then the order of the reaction is

a) Zero b) one

c) Fraction d) none

21. In a homogeneous reaction $A \longrightarrow B + C + D$, the initial pressure was P_0 and after time t it was P. expression for rate constant in terms of P_0, P and t will be

a) $k = \left(\frac{2.303}{t}\right)\log\left(\frac{2P_0}{3P_0 - P}\right)$

b) $k = \left(\frac{2.303}{t}\right)\log\left(\frac{2P_0}{P_0 - P}\right)$

c) $k = \left(\frac{2.303}{t}\right)\log\left(\frac{3P_0 - P}{2P_0}\right)$

d) $k = \left(\frac{2.303}{t}\right)\log\left(\frac{2P_0}{3P_0 - 2P}\right)$

22. If 75% of a first order reaction was completed in 60 minutes , 50% of the same reaction under the same conditions would be completed in

a) 20 minutes b) 30 minutes

c) 35 minutes d) 75 minutes

23. The half life period of a radioactive element is 140 days. After 560 days , 1 g of element will be reduced to

a) $\left(\frac{1}{2}\right)g$ b) $\left(\frac{1}{4}\right)g$

c) $\left(\frac{1}{8}\right)g$ d) $\left(\frac{1}{16}\right)g$

24. The correct difference between first and second order reactions is that (NEET)

a) A first order reaction can be catalysed; a second order reaction cannot be catalysed.

b) The half life of a first order reaction does not depend on $[A_0]$; the half life of a second order reaction does depend on $[A_0]$.

c) The rate of a first order reaction does not depend on reactant concentrations; the rate of a second order reaction does depend on reactant concentrations.

d) The rate of a first order reaction does depend on reactant concentrations; the rate of a second order reaction does not depend on reactant concentrations.

25. After 2 hours, a radioactive substance becomes $\left(\frac{1}{16}\right)^{th}$ of original amount.

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